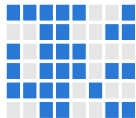
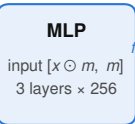


(a) Pre-training, once per dataset

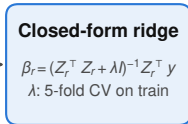
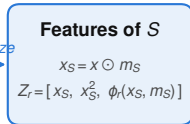
rows $\times$ all columns held,
random visibility mask m



■ visible ■ hidden



freeze



Objective

predict the hidden columns from the visible ones; keep last hidden layer as frozen features ϕ_r (seeds $r=1, 2, 3$)

Estimate

$\hat{y} = \frac{1}{3} \sum_r Z_r \beta_r$; $\hat{V}_y(S) = R_{\text{test}}^2(\hat{y}) + \text{closure}$
safeguards: feature s.d. floor 10^{-3} , clip to training range

(b) Query (S, y): subset-specific readout

(c) Scan–certify

